

Handling the Ionosphere in Solve

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Abstract

This note describes how Solve handles the effect of propagating through the ionosphere. This causes changes in the measured Delay and Rate.

1 Introduction

This document gives a brief introduction about how to model and handle the effect of the ionosphere. There are two general approaches:

1. Solve for the ionospheric effect by observing at two or more bands. This is the approach Solve uses in S/X sessions (or any session with two bands).
2. If single band observing, say K-band, we do not have enough information to solve for the ionosphere and must use information from some external source, for example GNSS data or a theoretical model. This situation is not routine and requires special code both to read in the external ionosphere data and to apply it to solve.

VGOS observing is a special case. In VGOS observing the the VLBI system observes at 4 bands each 512Mhz wide. (In the future these bands may be 1024MHz.) The ionosphere is solved for during correlation, and the correlator reports an ionospheric free Delay and Rate. Hence for these sessions you do not need to apply any ionosphere correction in Solve.

2 Physics

The ionosphere consists of the upper region of the atmosphere from roughly 80km to 580km where there is significant ionization and hence a large number of free electrons. The thickness of the ionosphere varies over time and depends on both the time of day and the level of solar activity. The presence of the free electrons causes the ionosphere to be dispersive—that is, different radio frequencies travel at different speeds, with lower frequencies traveling more slowly. This affects any technique that relies on radio signal traveling through the atmosphere. In particular it affects VLBI, GNSS (=GPS) and DORIS (Doppler Orbitography and Radiopositioning Integrated by Satellite [French system]).

Although this note focuses on VLBI the same methods apply to these other techniques.

In VLBI we work mostly with the Group Delay, and the discussion below will focus on this. There are other VLBI observables, such as Group Rate, Phase Delay or Single Band Delay, and the ionosphere also affects these. For simplicity in what follows I will focus on Group Delay but bear in mind that the effect is more general. Let τ_0 be the delay in the absence of the ionosphere, and $\tau_g(f)$ the frequency dependent group delay in the presence of the ionosphere. The equation relating the two is:

$$\tau_g(f) = \tau_0 + \tau_{ion}(f) \quad (1)$$

where $\tau_{ion}(f)$ is the delay caused by the ionosphere. This is positive because the ionosphere delays the signal. For the phase delay $\tau_p(f)$ the equation is similar, but the sign of the correction is opposite:

$$\tau_p(f) = \tau_0 - \tau_{ion}(f) \quad (2)$$

The ionospheric delay is determined by the number of free electrons traversed by the ray path. This number is called the Total Electron Content or TEC for short. It is conventionally measured in terms of electrons per cross-sectional area and the unit is called *TECU*.

$$1 \times TECU = 10^{16} \#el/m^2 \quad (3)$$

where $\#el$ is the number of electrons. Typical values of the ionosphere are 0-100 during the day, but can be much higher during periods of high solar activity. The ionospheric delay is:

$$\tau_{ion}(f) = \kappa \times TEC_{path} \times \frac{1}{f^2} \quad (4)$$

Here TEC_{path} is the *TEC* along the ray path. κ is a constant given by:

$$\kappa = \frac{c^2}{2\pi r_e^2} \simeq 40.3 m^3/s^2 \quad (5)$$

where r_e is the classical radius of the electron. This assumes that the delay is measured in meters. If the delay is measured in seconds you need to divide this by the speed of light.

TEC_{path} is the total number of electrons along the ray path. As a first approximation we can assume that this depends only on the elevation angle. Then we can relate $TEC(El)$ to the TEC in the zenith direction TEC_{Zenith} by a mapping function:

$$TEC(El) = TEC_{Zenith} \times Map_{TEC}(El) \simeq TEC_{Zenith} \frac{1}{\sin(El)} \quad (6)$$

If we are given the *TEC* then we can find the ionospheric delay. One option is to use the *TEC* from some external source, say from GNSS measurements.

The IGS provides TEC maps as a data product. This has been used in VLBI data analysis in some special circumstances. Since VLBI involves measuring the differential delay we must calculate:

$$\Delta TEC = TEC_{1,path} - TEC_{2,path} \quad (7)$$

where $TEC_{i,path}$ is the TEC along the ray path for antenna i .

3 Dual Band Measurements of the Ionosphere

The normal VLBI observable is the Group Delay τ_g , although the Phase Delay τ_p is used in some special circumstances. In the rest of this note we will work the Group Delay unless otherwise specified. In what follows we will omit the subscript g or p and instead use the subscript to indicate the frequency of the observation.

Combining Equations 1 and 4 we can rewrite the effect of the ionosphere as:

$$\tau_f = \tau_0 + \frac{\alpha}{f^2} \quad (8)$$

Here α depends on the TEC and the signal path but is independent of the frequency. Since this equation is linear in α , the value of α , and hence the ionospheric delay, can be found by measuring the delay at two frequencies. In GPS this is done by measuring at the two L band frequencies, L_1 and L_2 . In VLBI this is done by measuring the delay at S-band and X-band. The observed delay at these two frequencies is given by:

$$\tau_X = \tau_0 + \frac{\alpha}{f_X^2} \quad (9)$$

$$\tau_S = \tau_0 + \frac{\alpha}{f_S^2} \quad (10)$$

Multiplying the first of these by f_X^2 and the second by f_S^2 and subtracting we get:

$$\tau_X f_X^2 - \tau_S f_S^2 = \tau_0 \times (f_X^2 - f_S^2) \quad (11)$$

which can be solved for τ_0 :

$$\tau_0 = \tau_X \times \frac{f_X^2}{f_X^2 - f_S^2} - \tau_S \times \frac{f_S^2}{f_X^2 - f_S^2} \quad (12)$$

This linear combination is called ‘the ionospheric free delay’. This equation can also be written as:

$$\tau_0 = \tau_X + (\tau_X - \tau_S) \times \frac{f_S^2}{f_X^2 - f_S^2} \quad (13)$$

In this form the ionospheric delay τ_0 can be viewed as the measured X-band delay τ_X plus an ionospheric correction $\tau_{X,iono}$:

$$\tau_0 = \tau_X + \tau_{X,iono} \quad (14)$$

where

$$\tau_{X,iono} = (\tau_X - \tau_S) \times \frac{f_S^2}{f_X^2 - f_S^2} \quad (15)$$

We could also calculate the ionospheric delay for the S-band delay τ_S by exchanging S and X .

3.1 Aside: Uncertainty of a Derived Quantity

Suppose that we have a derived quantity $\tau_{Derived}$ that depends linearly on several measured quantities τ_i

$$\tau_{Derived} = \sum_i A_i \tau_i \quad (16)$$

The variance of $\tau_{Derived}$ is given by:

$$\sigma_{\tau_{Derived}}^2 = \sum_{i,j} A_i \sigma_i A_j \sigma_j C_{ij} \quad (17)$$

Where C_{ij} is the correlation matrix between the τ_i . If the τ_i are uncorrelated (that is, the measurements are independent) then this reduces to:

$$\sigma_{\tau_{Derived}}^2 = \sum_i A_i^2 \sigma_i^2 \quad (18)$$

3.2 Uncertainty of the Ionospheric Correction

The ionosphere correction is derived from two independent measured quantities τ_X and τ_S . Based on the the previous subsection, then

$$\sigma_{\tau_{X,iono}}^2 = (\sigma_{\tau_X}^2 + \sigma_{\tau_S}^2) \times \left(\frac{f_S^2}{f_X^2 - f_S^2} \right)^2 \quad (19)$$

hence

$$\sigma_{\tau_{X,iono}} = \sqrt{\sigma_{\tau_X}^2 + \sigma_{\tau_S}^2} \times \frac{f_S^2}{f_X^2 - f_S^2} \quad (20)$$

Note the following:

1. The uncertainty in the X-band ionosphere correction depends on the uncertainty of the measured delay in both bands added in quadrature times a scaling factor.
2. The scaling factor is frequency dependent and gets smaller the larger the difference in frequencies.

For typical values of S-band of 2GHz and X-band of 8Ghz we have

$$\frac{f_S^2}{f_X^2 - f_S^2} \simeq 1/15 \quad (21)$$

3.3 Uncertainty of Ionosphere Free Delay τ_0

In constructing the normal equations we need the uncertainty of the observables. In VLBI we work with the ionosphere free delay τ_0 which is composed of two quantities—the measured X-band delay τ_X and the ionosphere correction $\tau_{X,iono}$. Until roughly 2012 the calculation that Solve used was:

$$\sigma_{\tau_0,naive}^2 = \sigma_{\tau_X}^2 + \sigma_{\tau_{X,iono}}^2 \quad (22)$$

The subscript ‘*naive*’ indicates that this is not quite correct. Expanding the RHS of this we get:

$$\sigma_{\tau_0,naive}^2 = \sigma_{\tau_X}^2 + (\sigma_{\tau_X}^2 + \sigma_{\tau_S}^2) \times \left(\frac{f_S^2}{f_X^2 - f_S^2} \right)^2 \quad (23)$$

As indicated above, this is not quite correct. The problem is that τ_X and the ionosphere correction $\tau_{X,iono}$ are not independent. We can correct this by accounting for the correlation between the two but there is an easier way.

To find the correct uncertainty for τ_0 it is more useful to go back to the original Equation 12:

$$\tau_0 = \tau_X \times \frac{f_X^2}{f_X^2 - f_S^2} - \tau_S \times \frac{f_S^2}{f_X^2 - f_S^2} \quad (24)$$

In this case the variance in τ_0 is

$$\sigma_{\tau_0}^2 = \frac{\sigma_{\tau_X}^2 f_X^4 + \sigma_{\tau_S}^2 f_S^4}{(f_X^2 - f_S^2)^2} \quad (25)$$

If we take the difference between this and the naive value we find (after some simplification):

$$\sigma_{\tau_0}^2 - \sigma_{\tau_0,naive}^2 = 2\sigma_{\tau_X}^2 \frac{f_S^2}{f_X^2 - f_S^2} \quad (26)$$

Hence $\sigma_{\tau_0}^2 > \sigma_{\tau_0,naive}^2$: The naive value underestimates the uncertainty. There are alternative forms of the above equation that are useful.

$$\begin{aligned} \sigma_{\tau_0}^2 &= 2\sigma_{\tau_X}^2 \frac{f_S^2}{f_X^2 - f_S^2} + \sigma_{\tau_0,naive}^2 \\ &= \sigma_{\tau_X}^2 \times \left(1 + 2 \frac{f_S^2}{f_X^2 - f_S^2} \right) + \sigma_{\tau_{X,iono}}^2 \end{aligned} \quad (27)$$

We close this section by noting that Equation 22 is correct if the measured X-band delay is independent of the ionospheric delay. This will be the case if the ionospheric delay comes from some external source, say GNSS.

4 Ionosphere related nuSolve Calculations

By the time data is available for Solve processing the ionospheric correction has already been derived. This calculation is done by nuSolve which stores the results in the vgosDB. This section briefly describes what is calculated and where it is stored.

4.1 Kinds of Ionosphere Correction

VLBI has several different kinds of observables:

1. Group Delay. This is the most commonly used observable.
2. Group Rate. This is sometimes used, particularly if the Group Delay data is sparse or weak.
3. Single Band Delay. This is only used in preliminary processing and has large errors.
4. Phase Delay. This is only used in special situations and these solutions are called ‘Phase-Delay’ solutions.

The form for the first three of these is the same, while the sign of the last (Phase Delay) is opposite. The ionosphere correction depends on both the S- and X-band measurements. If one of these is missing or bad you cannot calculate the correction. NuSolve routinely calculates the Group Delay ionosphere correction. This correction can also be applied to Single Band Delay. NuSolve will only calculate the Phase Delay ionosphere correction if the analyst performs a Phase Delay solution. The X-band ionosphere correction, the uncertainty, and a flag indicating if the ionosphere correction is good is stored in the vgosDB file `ObsDerived/CalslntPathIonoGroup_X.nc`. If available the Phase Delay correction is stored in `ObsDerived/CalslntPathIonoPhase_X.nc`.

In principle you can apply the Group Delay ionosphere correction to the Phase Delay measurements or vice-versa. When you do so you must take care that you are using the appropriate sign.

4.2 What are the Appropriate Frequencies

In the above sections we frequently referred to f_S and f_X without defining what they are. In this section we give a fuller definition of this of.

In conventional VLBI we observe at two frequency bands: S-band which is around 2 GHz, and X-band which is around 8 GHz. In each of these we observe at several different ‘channels’. Within each session these (usually) stay the same, with typically 6 S-band channels and 10 X-band channels. Between sessions the number, frequency and bandwidth of the channels varies.

Since we observe at several different frequencies within each band an appropriate question is what frequencies we should use for the ionospheric calibration?

This problem was looked at by Tom Herring for his PhD Thesis which was published in the mid 1980s. NuSolve calculates ‘effective’ ionospheric frequencies for each band. These are functions of the channel frequencies and the strength of the signal in each channel. The calculation is slightly different for Group Delay, Group Rate and Phase Delay. The effective frequencies are stored in `EffFreq_bX.nc`.

5 Ionosphere Calculations in `get_calib`

In the previous sections we had a largely theoretical discussion about the ionosphere. In this section we discuss applying this to solve.

The ionospheric correction is applied in the routine `get_calib.f`. In the remainder of what follows we focus only on the ionosphere code. This code is complicated because of the following:

1. There are several types of ionosphere correction for different data types: Group Delay, Phase Delay and Group Rate. These have similar but slightly different forms.
2. As originally written, `get_calib` was used to apply the ionosphere correction for either S-band ($\simeq 2\text{Ghz}$) data or X-band data ($\simeq 8\text{Ghz}$). Hence part of the code tries to figure out which of these we are working with.
3. Some of the code tries to reconstruct reasonable values for missing data. Why is the data missing? Some of the earlier sessions are missing S-band related data. Storage space was expensive and the philosophy was to not keep data once it was used. Since S-band data is only used to calibrate the ionosphere it was discarded after this step. Unfortunately the part that calculates the uncertainty in the ionosphere free delay was incorrect.
4. For compatibility and testing, we wanted to preserve the ability to reproduce earlier incorrect calculations. Maintaining this code is no longer necessary.

The only information we are certain to have in `get_calib` is:

1. The Ionosphere correction is stored in `GION` as a two element array. The first is the Delay correction and the second the Rate correction.
2. The sigma of ionosphere correction is stored in `GION_SIG`
3. Effective frequency of current band is stored in `EFFREQ`. This is usually X-band in which case we have f_X .

Applying the ionospheric correction in `get_calib.f` is straightforward since we have it. Calculating the uncertainty of the ionosphere free observables is

more involved since some of the data required may be missing for older sessions. For use in what follows it is useful to repeat part of Equation 27:

$$\sigma_{\tau_0}^2 = \sigma_{\tau_X}^2 \times \left(1 + 2 \frac{f_S^2}{f_X^2 - f_S^2}\right) + \sigma_{X,iono}^2 \quad (28)$$

This gives the uncertainty in terms of the X-band band uncertainty, f_S^2 and f_X^2 and the uncertainty of the ionospheric calibration. For recent VLBI sessions all of this information is available. For older VLBI sessions the only thing that is missing is $\frac{f_S^2}{f_X^2 - f_S^2}$. A reasonable value for this based on current sessions is 0.079.

Table 1: Ionosphere Related variables in `get_calib`

Variable	Description	Default
MIN_FRQ	Parameter. Used to see if f_S is valid.	200
XS_FRQ	Parameter. Border between S and X-band	4000
GROUP_F_RATIO_SQ	Group Delay ionosphere scale factor	0.079
RATE_F_RATIO_SQ	Group Rate ionosphere scale factor	0.079
PHASE_F_RATIO_SQ	Phase Delay ionosphere scale factor	0.079
	Group Delay	
EffFreq	Effective freq. of current band, usually f_X	
EffFreq_XS	Effective freq. of other band, usually f_S .	
	Phase Delay	
PhEffFreq	Effective freq. of current band, usually f_X	
PhEffFreq_XS	Effective freq. of other band, usually f_S .	
	Group Rate	
REffFreq	Effective freq. of current band, usually f_X	
REffFreq_XS	Effective freq. of other band, usually f_S .	

Here is a an overview of how `get_calib.f` calculates the uncertainty of the ionosphere free observable. This is either the Delay or Rate or both.

1. Initialize $\frac{f_S^2}{f_X^2 - f_S^2}$ to the reasonable value 0.079. Store this in the variables `GROUP_F_RATIO_SQ`, `RATE_F_RATIO_SQ` and `PHASE_F_RATIO_SQ`
2. See if the current frequency is X-band or S-band. This is done by seeing if `EffFreq > XS_FRQ`.
3. See if the frequency stored in `EFFREQ_XS > MIN_FRQ`. This means that it is valid. If so then calculate the true values of `GROUP_F_RATIO_SQ`, `RATE_F_RATIO_SQ` and `PHASE_F_RATIO_SQ`.
4. Calculate $(1 + 2 \frac{f_S^2}{f_X^2 - f_S^2})$ which appears in the above equation. For the group delay this is stored in `TEMP_GROUP`. For the Phase Delay and Group Rate it is stored in similar variables.

5. Calculate the uncertainty in the Ionosphere Free Observable. Note that as written the code applies the Group Delay Correction to the Phase Delay.

In closing we note some subtleties about `get_calib`.

1. This routine calculates changes to the Theoretical Observable.
2. It has two slots, one for Delay and one for Rate.
3. There are several different Delay observables: Group Delay, Single Band Delay, and Phase Delay.
 - (a) Group Delay and Single Band Delay behave similarly with respect to the ionosphere, whereas for Phase Delay the sign is reserved.
 - (b) Because of this the theoretical delay changes differently for these two cases.
 - (c) Since there is only one slot for the Theoretical Delay we can't treat both Phase Delay and Group Delay (or Single Band Delay) at the same time.
4. You can apply the Group Delay ionosphere correction $\tau_{Group,X,ion}$ to Phase Delay $\tau_{Phase,X}$ or vice versa. However
 - (a) You need to use the appropriate sign of the correction—positive for Group Delay, negative for Phase Delay.
 - (b) In this case the ionospheric correction is independent of the Delay. This means that you don't need the extra term $2 \frac{f_S^2}{f_X^2 - f_S^2}$ in calculating the uncertainty.
 - (c) The reference frequencies for Group and Phase Delay are different. If you apply $\tau_{Group,X,ion}$ the Phase Delay you need to scale by $f_{X,group}^2 / f_{X,phase}^2$